How to Rule Out Periodic Orbits

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Idea: Periodic return versus monotone change

If $z(t)$ has period $T > 0$ and V is a single-valued function of state, then

$$V(z(T)) - V(z(0)) = 0.$$

Therefore either inequality

$$V(z(T)) - V(z(0)) < 0 \qquad \text{or} \qquad V(z(T)) - V(z(0)) > 0$$

rules out a nonconstant periodic orbit.

Definition: Gradient system

Let $U \subseteq \mathbb{R}^2$ be open. A system is a **gradient system** on U if there is a single-valued function $V \in C^1(U)$ such that

$$\dot{z} = -\nabla V(z).$$

The function V is called a **potential** for the system.

Observation: Potential decreases along a gradient flow

Along every solution of $\dot{z} = -\nabla V(z)$,

$$\dot{V} = \nabla V \cdot \dot{z} = -\|\nabla V\|^2 = -\|\dot{z}\|^2 \leq 0.$$

Recall: Simply connected planar domain

An open, path-connected set $U \subseteq \mathbb{R}^2$ is **simply connected** if every closed curve in U can be continuously contracted to a point while remaining in U.

Recall: Planar gradient test

Let $U \subseteq \mathbb{R}^2$ be open and let $F = (f, g) \in C^1(U)$. The identity

$$f_y = g_x$$

is necessary for $F = -\nabla V$. If U is simply connected, the identity is also sufficient: there is a function $V \in C^2(U)$ such that $F = -\nabla V$.

Theorem: Gradient flows have no closed orbit

A gradient system has no nonconstant periodic orbit.

Proof. Suppose $z(t)$ is T-periodic. Then

$$\begin{aligned} 0 &= V(z(T)) - V(z(0)) \\ &= \int_0^T \dot{V}(z(t))\,dt \\ &= -\int_0^T \|\dot{z}(t)\|^2\,dt. \end{aligned}$$

The continuous nonnegative integrand must vanish identically. Thus $\dot{z} \equiv 0$, so the solution is an equilibrium, not a nonconstant periodic orbit. $\square$

Question: Construct a potential

Show that

$$\dot{x} = \sin y, \qquad \dot{y} = x \cos y$$

has no closed orbit.

Solution. We seek V with

$$V_x = -\sin y, \qquad V_y = -x \cos y.$$

Integrating the first equation with respect to x gives

$$V(x, y) = -x \sin y + C(y).$$

Then

$$V_y = -x \cos y + C'(y).$$

The second equation requires $C'(y) = 0$, so we may take

$$V(x, y) = -x \sin y.$$

Hence $F = -\nabla V$, and the gradient-flow theorem rules out every nonconstant periodic orbit.

Definition: Strict Lyapunov function

Let z_* be an equilibrium of $\dot{z} = F(z)$ on an open set U. A function $V \in C^1(U)$ is a **strict Lyapunov function for z_* on U** if

$$V(z_*) = 0, \qquad V(z) > 0 \quad (z \neq z_*),$$

and

$$\dot{V}(z) = \nabla V(z) \cdot F(z) < 0 \quad (z \neq z_*).$$

Observation: Strict decrease forbids periodic return

If V is a strict Lyapunov function on U, then no nonconstant periodic orbit is contained in U.

Theorem: Strict Lyapunov stability criterion

Suppose F is locally Lipschitz on U and V is a strict Lyapunov function for z_* on a neighborhood U. If $c > 0$ is chosen so that

$$D_c = \{z \in U : V(z) \leq c\}$$

is compact and contains z_* in its interior, then z_* is asymptotically stable. More precisely, every solution with $z(0) \in D_c$ is defined for all $t \geq 0$, remains in D_c, and converges to z_*. Moreover, z_* is Lyapunov stable, and D_c is contained in its basin of attraction.

Corollary: Global strict Lyapunov criterion

Let F be locally Lipschitz on $\mathbb{R}^2$, let z_* be an equilibrium, and let V be a strict Lyapunov function for z_* on $\mathbb{R}^2$. If

$$V(z) \longrightarrow \infty \qquad (\|z\| \to \infty),$$

then every solution remains in the compact sublevel set determined by its initial value, extends to every $t \geq 0$, and converges to z_*. Thus z_* is globally asymptotically stable.

Question: Monotone energy without a gradient field

Show that

$$\ddot{x} + (\dot{x})^3 + x = 0$$

has no nonconstant periodic solution.

Solution. Set $y = \dot{x}$. Then

$$\dot{x} = y, \qquad \dot{y} = -x - y^3.$$

For

$$E(x, y) = \frac{1}{2}(x^2 + y^2),$$

we have

$$\dot{E} = x\dot{x} + y\dot{y} = xy + y(-x - y^3) = -y^4 \leq 0.$$

If a solution had period $T > 0$, then

$$0 = E(x(T), y(T)) - E(x(0), y(0)) = -\int_0^T y(t)^4 \, dt.$$

Hence $y \equiv 0$, and $\dot{y} = -x$ then forces $x \equiv 0$. Thus the only periodic solution is the equilibrium.

Question: Choose a Lyapunov function by canceling a mixed term

Analyze

$$\dot{x} = -x + 4y, \qquad \dot{y} = -x - y^3.$$

Solution. Let

$$V(x, y) = x^2 + ay^2, \qquad a > 0.$$

Then

$$\begin{aligned}\dot{V} &= 2x(-x + 4y) + 2ay(-x - y^3) \\ &= -2x^2 + (8 - 2a)xy - 2ay^4.\end{aligned}$$

Choose $a = 4$. Then

$$V = x^2 + 4y^2, \qquad \dot{V} = -2x^2 - 8y^4 < 0$$

away from $(0, 0)$. Moreover,

$$V(x, y) \longrightarrow \infty \qquad \left(\sqrt{x^2 + y^2} \to \infty\right).$$

Therefore every sublevel set $\{V \le c\}$, $c \ge 0$, is compact and positively invariant. Hence the origin is globally asymptotically stable and no nonconstant periodic orbit exists.

Definition: Dulac multiplier

Let $U \subseteq \mathbb{R}^2$ be open and let $F \in C^1(U, \mathbb{R}^2)$. A function $B \in C^1(U)$ is a **Dulac multiplier on** U if either

$$\nabla \cdot (BF) > 0 \quad \text{throughout } U$$

or

$$\nabla \cdot (BF) < 0 \quad \text{throughout } U.$$

Theorem: Bendixson–Dulac criterion

Let $U \subseteq \mathbb{R}^2$ be open and simply connected. Let $F \in C^1(U)$, and suppose there is a function $B \in C^1(U)$ such that

$$\nabla \cdot (BF) > 0$$

throughout U, or such that

$$\nabla \cdot (BF) < 0$$

throughout U. Then no nonconstant closed orbit lies entirely in U.

Proof. Suppose a closed orbit C lies in U, and let D be its interior. Simple connectivity gives $D \subset U$. If n is the outward unit normal, the planar divergence theorem gives

$$\iint_D \nabla \cdot (BF)\, dA = \oint_C BF \cdot n\, ds.$$

The right side is zero because F is tangent to its trajectory. The left side is nonzero because its integrand has a strict constant sign. This is a contradiction. $\square$

Question: A Dulac multiplier in the positive quadrant

Consider

$$\dot{x} = x(2 - x - y), \qquad \dot{y} = y(4x - x^2 - 3)$$

on

$$U = \{(x, y) : x > 0,\ y > 0\}.$$

Use $B(x, y) = 1/(xy)$ to rule out closed orbits in U.

Solution. We have

$$BF = \left(\frac{2 - x - y}{y}, \frac{4x - x^2 - 3}{x}\right).$$

Therefore

$$\begin{aligned}\nabla \cdot (BF) &= \frac{\partial}{\partial x}\left(\frac{2 - x - y}{y}\right) + \frac{\partial}{\partial y}\left(\frac{4x - x^2 - 3}{x}\right) \\ &= -\frac{1}{y} < 0.\end{aligned}$$

The domain U is open and simply connected, and $B, F \in C^1(U)$. Bendixson–Dulac therefore excludes every closed orbit contained in U. The multiplier is singular on the coordinate axes, so this calculation does not apply to a region containing either axis.

Question: An exponential Dulac multiplier

Show that

$$\dot{x} = y, \qquad \dot{y} = -x - y + x^2 + y^2$$

has no closed orbit in $\mathbb{R}^2$.

Solution. Let

$$B(x, y) = e^{-2x}.$$

Then

$$\begin{aligned}\nabla \cdot (BF) &= \frac{\partial}{\partial x}(e^{-2x} y) + \frac{\partial}{\partial y}\left(e^{-2x}(-x - y + x^2 + y^2)\right) \\ &= -2e^{-2x} y + e^{-2x}(-1 + 2y) \\ &= -e^{-2x} < 0.\end{aligned}$$

The plane is simply connected and B, F are continuously differentiable, so Bendixson–Dulac rules out every closed orbit.

Corollary: One-sign Dulac divergence forbids two nested periodic orbits

Let $U \subseteq \mathbb{R}^2$ be open, let $B, F \in C^1(U)$, and suppose $\nabla \cdot (BF)$ has a strict constant sign on U. Then U cannot contain two distinct periodic orbits C_1 and C_2 for which the compact annular region between C_1 and C_2 is contained in U.

Proof. Let $D \subset U$ be the compact annular region between two such orbits. The divergence theorem on D gives

$$\iint_D \nabla \cdot (BF)\, dA = \oint_{C_1} BF \cdot n\, ds + \oint_{C_2} BF \cdot n\, ds = 0,$$

because F is tangent to both boundary components. The area integral is nonzero, a contradiction. □

Visualization: Three periodic-orbit nonexistence criteria

structure	sufficient condition	contradiction
$\dot{z} = -\nabla V$	$\dot{V} = -\lVert \dot{z} \rVert^2$	$V(z(T)) - V(z(0)) < 0$
general flow	$\dot{V} < 0$	$V(z(T)) - V(z(0)) < 0$
planar flow	$\nabla\cdot(BF)$ has a strict sign on a simply connected domain	nonzero area flux =zero boundary flux

Observation: Failure of a criterion

Failure to find a Lyapunov function V or a Dulac multiplier B gives no conclusion about whether a periodic orbit exists.